Development of a GPU Accelerated Fluid Structure Interaction Solver Using Immersed Boundary Methods on a Structured Mesh

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1 Extended Abstract

A GPU (Graphics Processing Unit) accelerated Fluid Structure Interaction Solver is proposed in this work. 3-D Incompressible Navier Stokes Equations are solved using a 2-step Fractional Step Method on Cartesian non body conformal meshes [3], while solid-fluid coupling is ensured using a diffused interface immersed boundary method [9]. The resulting linear equations are solved using parallelized preconditioned Conjugate-Gradient Algorithm. The results of the Lid Driven Cavity problem and 2-D flow around a square cylinder are presented in the next section, with a qualitative comparison with cited references.

The solver developed makes the use of a diffused interface type immersed boundary method [4] . The Immersed boundary methods, as initially introduced by Peskin in 1973 [2], are capable of solving fluid structure interaction flows without a body conformal mesh, for any complex curvilinear body. This gives one the flexibility to choose simple structured cartesian meshes over highly unstructured meshes, thus simplifying the grid generation task and preserving the accuracy of the discretization algorithm at the same time. Projection Immersed boundary approach [9], a variant of the Diffused interface direct forcing method, was chosen. This approach is perhaps easier to implement for simulation of moving boundaries, and exhibits a high degree of parallelism. The entire algorithm is implemented using the Single Instruction Multiple Thread (SIMT) execution model of GPU, thus accelerating the overall computations [10]. The subsequent linear equations resulting from the discretization are solved using a Preconditioned Conjugate Gradient Algorithm. Preconditioning is done using a Red-Black Symmetric Gauss Siedel Algorithm, owing to its inherent parallelism [6]. Thus the convergence is greatly accelerated by solving multiple nodes parallely using GPU architecture.

The objective is to develop a fully parallelized solver capable of solving flows around three dimensional complex bodies using GPU architecture, with minimal computational and human effort. A directive based parallel programming has been implemented using the open source OpenACC library.

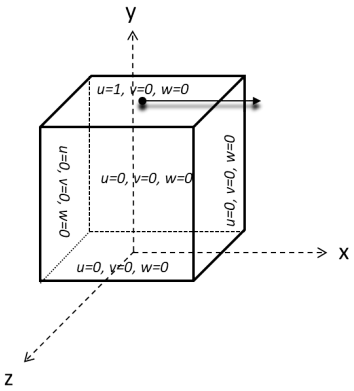


Fig. 1: Lid Driven Cavity Problem Description

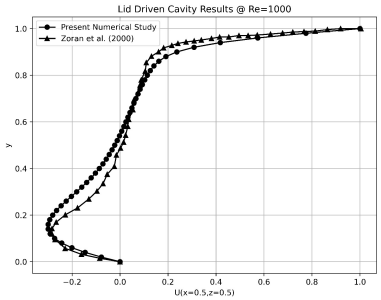


Fig. 2: Transverse variation of x component of velocity along the centerline on the x-y midplane compared with the results of [8]

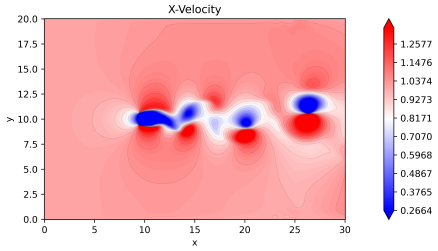


Fig. 3: Contour plot of x-component of velocity for flow around square cylinder

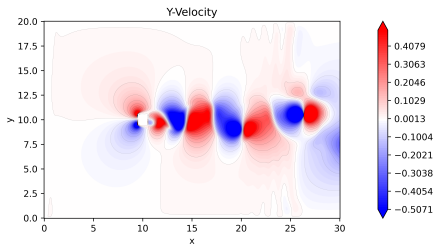


Fig. 4: Contour plot of y-component of velocity for flow around square cylinder

2 Preliminary Results

The standard Lid Driven Cavity problem, as described in Fig.1 was solved to investigate the accuracy of the solver in 3-D domain. A 50×50×50 Non Uniform Mesh was used. The velocity along the x-direction at the midplane (x-y) was compared with 3D Lid driven Cavity results of [8] at a Reynold number of 1000, as shown in Fig. 2. The overall simulation took 6000 seconds to converge, for a physical time of 100s and time step Δt=0.005s. Further, to validate the implementation of the immersed boundary method, flow around a square cylinder was simulated at various reynold numbers. Fig. 3 and Fig. 4 show the x-velocity contours and the y-velocity contours, respectively, at Reynold number equal to 150. Vortex shedding patterns can be clearly observed. These simulations were carried out sequentially on an Intel i-7 7th generation 16-core CPU. Parallelization of the flow solver algorithm on GPU is underway, and is expected to greatly reduce the computational efforts as compared to the conventional solvers.

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